SQL CASE Statement

Exercize 3

1. The products Table

```
SELECT  product_name,
        Price,
CASE    WHEN price > 1000 THEN 'expensive'
        WHEN price BETWEEN 100 AND 1000 THEN 'mid_range'
        WHEN price < 100 THEN 'budget'
END AS  price_Category
FROM    products;
```

Products

Product_name	Price	Price_category
Laptop	1200.00	expensive
Phone	800.00	mid_range
Keyboard	45.00	budget
Monitor	300.00	mid_range
Mouse	25.00	budget

2. The Orders Table

```
SELECT  Customer_name,
        amount,
CASE    WHEN amount >= 1000 THEN 'high_value'
        WHEN amount BETWEEN 500 AND 999.99 THEN 'med_value'
        WHEN amount < 500 THEN 'low_value'
END AS  Order_value_category
FROM    Orders
```

Orders

Customer name	amount	Order_value_category
Alice	150.00	low value
Bob	560.00	med value
Charlie	999.99	med value
Diana	45.50	low value
Ethan	1200.00	high value

3. The employees Table

```
SELECT   emp_name,
         department,
         Salary,
    CASE WHEN department = 'IT' AND Salary >80000 THEN
         'Senior IT'
         WHEN department = 'HR' AND salary >55000 THEN
         'experienced HR'
         ELSE 'Staff'
    END AS position_level
    FROM employees;
```

employees

emp_name	department	Salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4. The Students Table

```
SELECT Student_name,
       Score,
       grade,
  CASE WHEN score >= 90 THEN 'A'
       WHEN score >= 80 THEN 'B'
       WHEN Score >= 70 THEN 'C'
       WHEN Score >= 60 THEN 'D'
       ELSE 'F'
  END As grade
  FROM Students;
```

Students

Student_name	Score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5. The deliveries Table

```
SELECT delivery_id,
       delivery_time_minutes,
       performance,
  CASE WHEN time_minutes <= 30 THEN 'Fast'
       WHEN time_minutes BETWEEN 31 AND 60 THEN 'ontime'
       WHEN time_minutes > 60 THEN 'late'
  END AS performance
  FROM deliveries;
```

delivery_id	time_minutes	performance
1	45	On time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

6. The tickets Table

```
SELECT issue_type,
       Priority,
  CASE WHEN priority = 3 THEN 'high'
       WHEN priority = 2 THEN 'medium'
       WHEN priority = 1 THEN 'low'
  END AS Priority_level
  FROM tickets;
```

tickets

Issue_type	Priority	Priority_level
login issue	1	Low
Server down	3	high
Slow system	2	medium
Email error	2	medium
password rest	1	low

7. The attendance Table:

```
SELECT  student_id,
        (days_present * 100.0 / total_days) AS attendance_percentage,
CASE    WHEN (days_present *100 / total_days) >= 90 THEN 'Excellent'
        WHEN (days_present*100/ total_days) >= 75 THEN 'Good'
ELSE    'Needs Improvement'
```

```
END AS attendance_status
FROM attendance;
```

attendance

Student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	excellent.

8. The Products_Inventory Table

```
SELECT product_id,
       stock_qty,
CASE WHEN stock_qty = 0 THEN 'Out_of_Stock'
     WHEN               THEN 'low_stock'
     WHEN stock_qty > 5 THEN 'In_stock'
     WHEN stock_qty BETWEEN 1 AND 5 THEN 'low_stock'
```

Product_inventory

Product_id	stock_qty	stock status
1	5	low stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	low stock

9. The Classes Table

```
SELECT  Subject,
        enrolled_students,
CASE    WHEN enrolled_students >= 25 THEN 'large'
        WHEN enrolled_students BETWEEN 10 AND 24 THEN 'medium'
        WHEN enrolled_students < 10 THEN 'Small'
AND AS  Class_Size_Category
FROM    Classes
```

Classes ~~enrolled_students~~

Subject	enrolled_students ~~30~~	Class_size_Category
Math	~~25~~ 30	large
English	~~15~~ 25	large
Science	15	Small
Art	5	Small
History	20	medium

10.

```
SELECT  payment_id
        Payment_method,
        amount,
CASE    WHEN payment_method = 'Cash' AND amount >= 200
        THEN 'eligible for discount'
        ELSE 'not_eligible'
END AS  discount_eligibility
  FROM  payments;
```

Payments

payment_id	payment method	amount	discount_eligibility
1	Card	50.00	not eligible
2	Cash	200.00	eligible for discount
3	Card	150.00	not eligible
4	Paypal	7500	not eligible
5	Cash	300.00	eligible for discount.